Instruction to Generate Degenerate ETPA Figures using Matlab code

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Instructions for using the Matlab codes to generate the degenerate ETPA figures

1. Place all the provided Matlab codes and the .mat file (output of transition_dipole.py) in the same directory.
2. In the individual_etpa_fig_degenerate.m code
   - Change the variable in line 3 to match the name of the .mat file that you are using (without the extension)
   - Lines 5-17 are where the parameters that we enter are set.
     - Energies should be entered in eV
       - it is possible to set up for other units but the conversion factor I have set up later in code uses eV
     - Entanglement time should be in seconds
3. Run the individual_etpa_fig_degenerate.m code
   - This will generate the degenerate ETPA figure
     - Note that I only set up this figure to get a quick look at the results and have made other codes that generate good looking figures
       - See Combo_Figures for the codes that generate better looking figures
4. Outputs the Rgrid_symmetric_per file
   - Contains the degenerate ETPA spectra data (First column is energy values in eV, second column is absorption intensity)
5. Note that the Fortran code will also produce this file (named Rgrid_deg)
   - See the Fortran section of the Nondegenerate_ETPA_Figure for more info on Fortran code
   - Note that I was having some issues with the Fortran output of this file. I have fixed what I believe was the issue but haven't actually tested the new results due to this data taking 1 minute to generate using the Matlab code.